EE3621: Digital Signal Processing

Lecture 2: LTI Systems, Convolution, Stability & Difference Equations
(III B.Tech EEE)

Lecture Notes (40-Minute Plan)

1 Lecture Plan & Objectives (40 Minutes Breakdown)

- **00:00 – 05:00 (5 mins):** Review of System Properties & Introduction to LTI Systems.
- **05:00 – 15:00 (10 mins):** Derivation of the Convolution Sum & Graphical Steps.
- **15:00 – 22:00 (7 mins):** Properties of Discrete-Time Convolution (with proofs).
- **22:00 – 30:00 (8 mins):** LTI Causality, BIBO Stability (with proof), and Step Response.
- **30:00 – 38:00 (8 mins):** Constant-Coefficient Difference Equations: Homogeneous & Particular Solutions.
- **38:00 – 40:00 (2 mins):** Checkpoints & Quick Q&A.

2 Derivation of the Convolution Sum

An LTI system is completely characterized by its impulse response $h[n] = \mathcal{T}\{\delta[n]\}$. By the sifting property, any arbitrary input $x[n]$ is expressed as a weighted sum of impulses:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k] \tag{1}$$

Applying the system operator $\mathcal{T}$ to both sides:

$$y[n] = \mathcal{T}\{x[n]\} = \mathcal{T}\left\{\sum_{k=-\infty}^{\infty} x[k]\delta[n-k]\right\} \tag{2}$$

By the property of **Linearity** (superposition and scaling), we can move the operator inside the summation:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\mathcal{T}\{\delta[n-k]\} \tag{3}$$

By the property of **Time-Invariance**, if the response to a unit impulse $\delta[n]$ is $h[n]$, the response to a shifted impulse $\delta[n-k]$ is a shifted impulse response $h[n-k]$:

$$\mathcal{T}\{\delta[n-k]\} = h[n-k] \tag{4}$$

Substituting this back yields the **Convolution Sum**:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = x[n] * h[n] \tag{5}$$

This is the convolution sum, illustrated in Figure ??.

Figure 1: Step-by-step graphical progression of convolution: folding, shifting, multiplication, and summation.

3 Properties of Discrete-Time Convolution

3.1 Commutativity

$$x[n] * h[n] = h[n] * x[n] \tag{6}$$

Proof: Let $m = n - k \Rightarrow k = n - m$. As $k \to \infty, m \to -\infty$, and as $k \to -\infty, m \to \infty$:

$$x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = \sum_{m=-\infty}^{\infty} x[n-m]h[m] = \sum_{m=-\infty}^{\infty} h[m]x[n-m] = h[n] * x[n] \tag{7}$$

3.2 Associativity and Distributivity

$$(x[n] * h_1[n]) * h_2[n] = x[n] * (h_1[n] * h_2[n]) \tag{8}$$
$$x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n] \tag{9}$$

3.3 Shifting Property

If $y[n] = x[n] * h[n]$, then:

$$x[n - n_1] * h[n - n_2] = y[n - n_1 - n_2] \tag{10}$$

3.4 Complex Exponentials as Eigenfunctions (Frequency Response Theorem)

A complex exponential sequence $x[n] = e^{j\omega n}$ is an **eigenfunction** of any LTI system. That is, convolving it with an impulse response $h[n]$ yields:

$$e^{j\omega n} * h[n] = H(e^{j\omega})e^{j\omega n} \tag{11}$$

where $H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k}$ is the frequency response of the system.
Proof:

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]e^{j\omega(n-k)} = \left(\sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k} \right) e^{j\omega n} = H(e^{j\omega})e^{j\omega n} \tag{12}$$

3.5 Duality: Complex Exponential System as a Frequency Extractor

If the system itself is a complex oscillator with impulse response $h[n] = e^{j\omega_0 n}$, then convolving it with an arbitrary input sequence $x[n]$ extracts the frequency component of $x[n]$ at ω_0:

$$x[n] * e^{j\omega_0 n} = X(e^{j\omega_0})e^{j\omega_0 n} \tag{13}$$

where $X(e^{j\omega_0}) = \sum_{k=-\infty}^{\infty} x[k]e^{-j\omega_0 k}$ is the DTFT of the input.
Proof:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]e^{j\omega_0(n-k)} = \left(\sum_{k=-\infty}^{\infty} x[k]e^{-j\omega_0 k} \right) e^{j\omega_0 n} = X(e^{j\omega_0})e^{j\omega_0 n} \tag{14}$$

At $n = 0$, the output is exactly $y[0] = X(e^{j\omega_0})$, representing the frequency component at ω_0.

4 Causality, BIBO Stability & Step Response

4.1 Causality

An LTI system is causal if and only if:

$$h[n] = 0 \quad \forall n < 0 \tag{15}$$

For a causal system and causal input, the convolution simplifies to:

$$y[n] = \sum_{k=0}^{n} x[k]h[n-k] \tag{16}$$

4.2 BIBO Stability

An LTI system is BIBO stable if and only if its impulse response is absolutely summable:

$$S = \sum_{n=-\infty}^{\infty} |h[n]| < \infty \tag{17}$$

Proof (Sufficiency): If the input is bounded, i.e., $|x[n]| \le M_x < \infty$, then:

$$|y[n]| = \left| \sum_{k=-\infty}^{\infty} x[n-k]h[k] \right| \le \sum_{k=-\infty}^{\infty} |x[n-k]||h[k]| \le M_x \sum_{k=-\infty}^{\infty} |h[k]| = M_x S < \infty \tag{18}$$

Since $S < \infty$, the output is bounded. Stability profiles are shown in Figure ??.

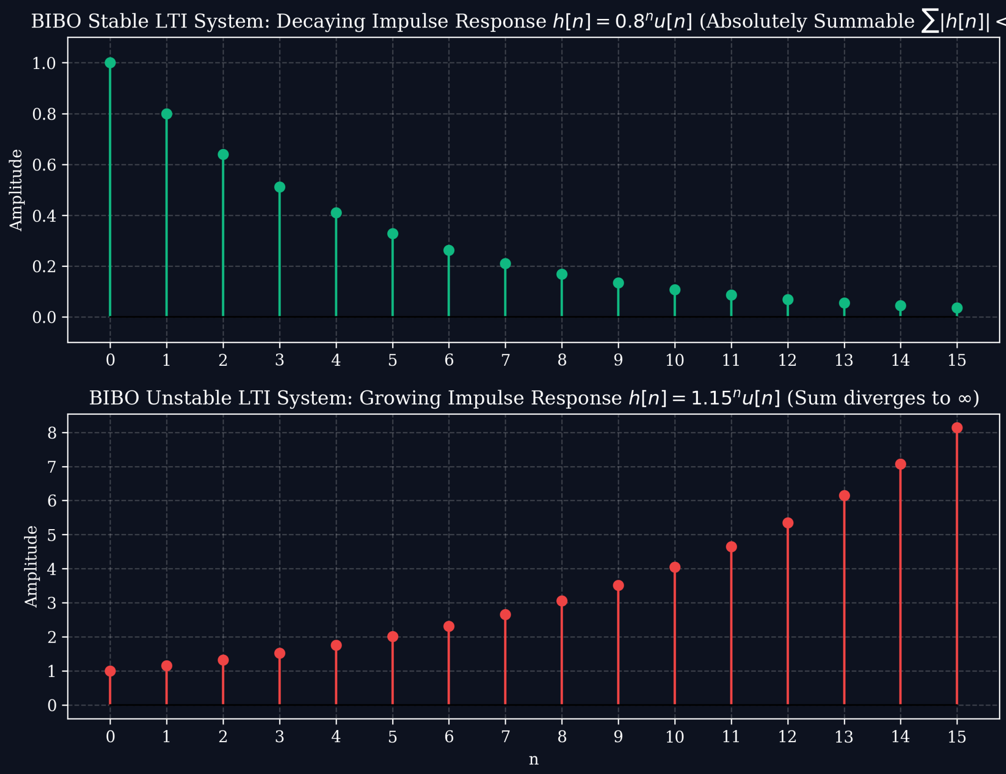

Figure 2: Impulse response profiles for stable and unstable discrete systems.

4.3 Step Response

The unit step response is defined as:

$$s[n] = h[n] * u[n] = \sum_{k=-\infty}^{n} h[k] \tag{19}$$

The impulse response is recovered by:

$$h[n] = s[n] - s[n-1] \tag{20}$$

5 Constant-Coefficient Difference Equations

Discrete-time LTI recursive systems are described by difference equations:

$$y[n] + \sum_{k=1}^{N} a_k y[n-k] = \sum_{m=0}^{M} b_m x[n-m] \tag{21}$$

5.1 Note on Sign Conventions & Pole Stability

A common source of confusion in LTI difference equations is the sign convention of the feedback coefficients and how it relates to poles:

1. **Left-Hand Side (LHS) Form:** If we write the first-order system with all terms on the LHS:
$$y[n] + a_1 y[n-1] = x[n] \tag{22}$$
Here, the coefficient is $+a_1$. The pole is at $z = -a_1$. For stability, we require the pole magnitude to be less than 1: $|-a_1| < 1 \implies |a_1| < 1$.

2. **Right-Hand Side (RHS) Form (Feedback Form):** If we express the system with feedback on the RHS:
$$y[n] = ry[n-1] + x[n] \tag{23}$$
Here, the feedback coefficient is r (which corresponds to $-a_1$ in the LHS form). The pole is located directly at $z = r$. For stability, we require $|r| < 1$.

Stability does not restrict the feedback coefficient to being strictly positive or negative; it only requires the feedback magnitude to be less than one ($|r| < 1$ or $|a_1| < 1$). For example, both $y[n] = 0.8y[n-1] + x[n]$ (pole at 0.8) and $y[n] = -0.8y[n-1] + x[n]$ (pole at -0.8) are stable.

The total solution is $y[n] = y_h[n] + y_p[n]$.

5.2 Homogeneous Solution $y_h[n]$

Set $x[n] = 0$, yielding the homogeneous difference equation. Assume $y_h[n] = C\lambda^n$:

$$\lambda^N + a_1\lambda^{N-1} + \cdots + a_N = 0 \tag{24}$$

For distinct roots λ_k, the homogeneous solution is:

$$y_h[n] = \sum_{k=1}^{N} C_k \lambda_k^n \tag{25}$$

5.3 Particular Solution $y_p[n]$

Represented by trial functions corresponding to the inputs, e.g., $y_p[n] = K_p$ for constant inputs, or $y_p[n] = K_p\beta^n$ for exponential inputs $x[n] = A\beta^n$.

5.4 Analytical Solution Concrete Walkthrough

Solve the difference equation $y[n] - 0.5y[n-1] = x[n]$ for $x[n] = 2^n u[n]$ under initial rest conditions ($y[-1] = 0$).

1. **Homogeneous Solution $y_h[n]$:** Set input $x[n] = 0$. Substitute $y_h[n] = C\lambda^n$:
$$C\lambda^n - 0.5C\lambda^{n-1} = 0 \Rightarrow \lambda - 0.5 = 0 \Rightarrow \lambda = 0.5 \tag{26}$$
Thus, the natural response shape is:
$$y_h[n] = C(0.5)^n \tag{27}$$

2. **Particular Solution $y_p[n]$:** Since input is $x[n] = 2^n u[n]$, assume $y_p[n] = K \cdot 2^n$ for $n \geq 0$. Substitute into difference equation:

$$K \cdot 2^n - 0.5K \cdot 2^{n-1} = 2^n \tag{28}$$

Divide by 2^{n-1} to solve for K:

$$2K - 0.5K = 2 \Rightarrow 1.5K = 2 \Rightarrow K = \frac{4}{3} \tag{29}$$

Thus, $y_p[n] = \frac{4}{3}2^n$ for $n \geq 0$.

3. **Complete Solution $y[n]$:** Combine the components:

$$y[n] = C(0.5)^n + \frac{4}{3}2^n \quad (n \geq 0) \tag{30}$$

4. **Solve for Constant C:** Under initial rest $y[-1] = 0$. Direct evaluation of difference equation at $n = 0$:

$$y[0] = 0.5y[-1] + x[0] = 0.5(0) + 2^0 = 1 \tag{31}$$

Substitute $n = 0$ into complete solution:

$$y[0] = C + \frac{4}{3} \Rightarrow C + \frac{4}{3} = 1 \Rightarrow C = -\frac{1}{3} \tag{32}$$

Thus, the final complete solution is:

$$y[n] = \left[-\frac{1}{3}(0.5)^n + \frac{4}{3}2^n\right] u[n] \tag{33}$$

6 Hardware Implementation: Parallelization & Systolic MAC Arrays

In real-time digital signal processing systems (FPGAs, ASICs, and dedicated DSP processors), computing convolution sequentially is often too slow. Instead, the computation is parallelized using a **Systolic MAC (Multiply-Accumulate) Array** in the **Transposed Direct Form FIR** structure. This reduces the time complexity from $O(L_x \cdot L_h)$ to $O(L_x)$ overall, yielding $O(1)$ output throughput (one output sample per clock cycle).

6.1 Transposed Direct Form FIR Structure

Instead of storing input history and summing them via a large adder tree, the input sample $x[n]$ is broadcast to all processing cells simultaneously. Registers are placed along the accumulation path.

For a 3-tap ($L_h = 3$) filter:

$$\begin{array}{cccccccccc}
\text{Broadcast Input } x[n] & \rightarrow & \downarrow & & & \downarrow & & & \downarrow \\
 & & \boxed{\times h[0]} & & & \boxed{\times h[1]} & & & \boxed{\times h[2]} \\
 & & \downarrow & & & \downarrow & & & \downarrow \\
y[n] & \leftarrow & \boxed{+} \leftarrow & \boxed{\text{Reg } R_0} & \leftarrow & \boxed{+} \leftarrow & \boxed{\text{Reg } R_1} & \leftarrow & \boxed{\text{Reg } R_2}
\end{array}$$

6.2 Register Update Equations

At each clock cycle n, all registers update simultaneously based on the previous cycle's states:

$$R_0[n] = R_1[n-1] + x[n] \cdot h[0] \tag{34}$$
$$R_1[n] = R_2[n-1] + x[n] \cdot h[1] \tag{35}$$
$$R_2[n] = x[n] \cdot h[2] \tag{36}$$

The output is read directly from the accumulator: $y[n] = R_0[n]$. For $x[n] = \{1, 2, 1\}$ convolved with $h[n] = \{1, 2, 1\}$, the systolic execution matches the algebraic convolution $y[n] = \{1, 4, 6, 4, 1\}$.

7 Detailed Worked Numerical Examples

7.1 Example 1: Convolution of Finite Sequences

Find the convolution $y[n] = x[n] * h[n]$ for $x[n] = \{1, \mathbf{2}, 1\}$ and $h[n] = \{\mathbf{1}, 1\}$.

- $x[n]$ range: $n \in [-1, 1]$, length $L_x = 3$.
- $h[n]$ range: $n \in [0, 1]$, length $L_h = 2$.
- Output range: starts at $-1 + 0 = -1$, ends at $1 + 1 = 2$, length $L_y = 3 + 2 - 1 = 4$.

Calculating index-by-index:

$$\begin{aligned}
y[-1] &= x[-1]h[0] = 1 \cdot 1 = 1 \\
y[0] &= x[-1]h[1] + x[0]h[0] = 1 \cdot 1 + 2 \cdot 1 = 3 \\
y[1] &= x[0]h[1] + x[1]h[0] = 2 \cdot 1 + 1 \cdot 1 = 3 \\
y[2] &= x[1]h[1] = 1 \cdot 1 = 1
\end{aligned}$$

Result: $y[n] = \{1, \mathbf{3}, 3, 1\}$.

7.2 Example 2: Convolution of Causal Exponential Sequences

Compute $y[n] = x[n] * h[n]$ for $x[n] = a^n u[n]$ and $h[n] = b^n u[n]$ ($a \neq b$).

$$y[n] = \sum_{k=0}^{n} a^k b^{n-k} = b^n \sum_{k=0}^{n} (a/b)^k = b^n \left(\frac{1 - (a/b)^{n+1}}{1 - a/b} \right) = \frac{b^{n+1} - a^{n+1}}{b - a} u[n] = \frac{a^{n+1} - b^{n+1}}{a - b} u[n]$$

7.3 Example 3: Difference Equation Homogeneous and Particular Solutions

Solve $y[n] - 0.5y[n-1] = u[n]$ under initial rest conditions.

1. Homogeneous solution: $y_h[n] = C(0.5)^n$.
2. Particular solution: Try $y_p[n] = K \Rightarrow K - 0.5K = 1 \Rightarrow K = 2$.
3. Complete solution: $y[n] = C(0.5)^n + 2$ for $n \geq 0$.
4. Apply initial conditions: under initial rest, $y[-1] = 0 \Rightarrow y[0] = 0.5(0) + 1 = 1$.
5. At $n = 0$: $y[0] = C + 2 = 1 \Rightarrow C = -1$.
6. Result: $y[n] = (2 - 0.5^n)u[n]$.

7.4 Example 4: Stability Test

Prove if $h[n] = (0.9)^n u[n]$ is stable.

$$S = \sum_{n=0}^{\infty} |0.9|^n = \frac{1}{1 - 0.9} = 10 < \infty \quad \Longrightarrow \text{System is stable.}$$

7.5 Example 5: Step Response calculation

Calculate the step response $s[n]$ of a system with impulse response $h[n] = \delta[n] - \delta[n-1]$.

$$s[n] = \sum_{k=-\infty}^{n} (\delta[k] - \delta[k-1]) = u[n] - u[n-1] = \delta[n]$$